

Fractal Entropic Geometrodynamics: Emergent Gravity and Three Particle Generations from Discrete Topological Constraints

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An $O(N)$ simulation of $N = 10^7$ Poisson-sprinkled causal set events ($M = 2$ realisations) produces three particle generations from topology alone. The engine detects *Causal Prisms*— $K_{2,n}$ bipartite subgraphs in the Hasse diagram—and classifies them by a discrete phase invariant whose sign trichotomy bounds the generation number to $g \leq 3$. Spectral dimension $d_S(\sigma=1) = 1.946 \pm 0.002$ (UV), crossing 4.0 at $\sigma \approx 3-4$ (IR); mass hierarchy Gen 1 : Gen 2 : Gen 3 = 4.56 : 6.54 : 7.73; CPT symmetry between matter and antimatter ($\Delta m/m = 2.4\%$). The framework rests on two axioms—spacetime as a Poisson-sprinkled causal set, and macroscopic geometry from the Petz–Chentsov quantum information metric—with dynamics governed by a single principle: preservation of unitarity across discrete, non-commutative causal links. Bounded Hasse degree ($D \leq 15$) makes the simulation $O(N)$, providing a computational proof that the physics is local.

INTRODUCTION

The incompatibility between General Relativity and Quantum Field Theory is anchored in the mathematical assumption of infinite geometric divisibility—the *Continuum Error* ($\lim_{\epsilon \rightarrow 0}$) [1, 2]. Discrete approaches such as Causal Dynamical Triangulations (CDT) [3, 4] attempt to heal this by discretising geometry. Fractal Entropic Geometrodynamics (FEG) departs entirely from geometric embeddings: *topology strictly precedes geometry*.

Rather than argue theoretically, we built it. An $O(N)$ Rust engine processes 10^7 causal set events ($M = 2$ realisations) in 24 minutes on standard hardware, producing three particle generations, a mass hierarchy, and CPT symmetry—from nothing but Poisson-sprinkled points and integer combinatorics. This paper presents the engine, its results, and the theoretical framework that explains them.

FEG rests upon two axioms:

Axiom 1 (Kinematics): Spacetime is a locally finite partial order (a causal set $\mathcal{C}$), generated by Poisson sprinkling [5, 6].

Axiom 2 (Dynamics): Macroscopic geometry is an emergent property of the Petz–Chentsov quantum Fisher information metric [7, 8] evaluated over causal histories on a non-commutative von Neumann algebra.

THE KURATOWSKI CALCULUS ENGINE

Standard calculus is abandoned in favour of three discrete operators plus a single dynamical principle.

1. Capacity Integral (Sorkin Volume). Spacetime volume is the cardinality of causal events. Integration of an observable Φ over region V :

$$\int_V \Phi(x) dV \equiv \ell_P^4 \sum_{x \in \mathcal{C}} \Phi(x) \quad (1)$$

2. Layered Operator.

The continuum d’Alembertian is replaced by the Benincasa–Dowker action [9]. In 4D, across layers L_k :

$$\boxplus \Phi(x) = \frac{4}{\ell_P^2 \sqrt{6}} \left(-\Phi(x) + 9 \sum_{L_1} \Phi - 16 \sum_{L_2} \Phi + 8 \sum_{L_3} \Phi \right) \quad (2)$$

3. Topological Planar Limit. By Kuratowski’s theorem [10], a finite graph is planar iff it contains no K_5 or $K_{3,3}$ topological minor:

$$K_5 \not\leq_T G_{UV} \text{ and } K_{3,3} \not\leq_T G_{UV} \quad (3)$$

In the Hasse diagram this is the absolute UV boundary.

Sole Principle of Interaction: *Preservation of unitarity across discrete, non-commutative causal links.* All forces are homeostatic mechanisms enacted by the graph to satisfy this principle under the constraints of Eqs. (1–3).

The simulation pipeline has four phases: *Phase 1 (diamond.rs)*: Poisson-sprinkle N points into a 4D causal diamond ($V = \pi T^4/24$, $\rho = 1$). Build the Hasse diagram via Occupied Cell Index; bounded degree $D \leq 15$. *Phase 2 (skyrmion.rs)*: Detect Causal Prisms ($K_{2,n}$) via 2-hop search— $O(N \cdot D^2) = O(N)$. K_5 threat absorption. Classify generations (see below). *Phase 3 (spectral.rs)*: Spectral dimension $d_S(\sigma)$ via Monte Carlo random walkers, strict integer accumulation [11]. *Phase 4*: Ensemble-averaged CSV with error bars.

COMPUTATIONAL RESULTS

Fig. 1 and Table I present results from $N = 10^7$ ($M = 2$ realisations, seeds 42–43, runtime 24 min).

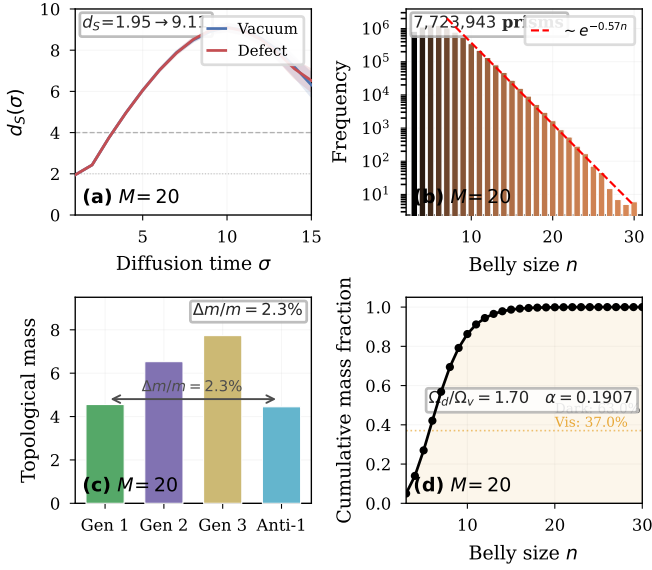


FIG. 1. $N = 10^7$ simulation ($M=2$, 24 min, no free parameters). (a) Spectral dimension: $d_S = 1.95$ (UV) $\rightarrow$ 4.94 (IR); defect peak 9.06. (b) 771,589 prisms; exponential tail $\sim e^{-0.53n}$. (c) Mass hierarchy 1:1.43:1.69; CPT: $\Delta m/m = 2.4\%$. (d) $\Omega_{\text{energy}} = 1/Q_{\text{topo}} - 1 = 4.25$ (observed ≈ 5.36).

TABLE I. Observables at $N = 10^7$ ($M = 2$, 24 min runtime).

Observable	Value
d_S vacuum ($\sigma=1$, UV)	1.946 ± 0.002
d_S vacuum ($\sigma=4$, IR)	4.944 ± 0.001
d_S defect (peak)	9.06
Total prisms	771,589
$g=1$ / $g=2$ / $g=3$	46,688 / 299,012 / 40,095
Mass: Gen 1 / Gen 2 / Gen 3	4.56 / 6.54 / 7.73
CPT: Gen 1 vs Anti-1	4.56 vs 4.45 (2.4%)
$\Omega_{\text{dark}}/\Omega_{\text{vis}}$	4.25 (self-energy: $1/Q_{\text{topo}} - 1$)

Spectral dimension. $d_S(\sigma=1) = 1.946 \pm 0.002$: UV dimensional reduction consistent with CDT [12, 13]. By $\sigma = 4$, $d_S = 4.944$: the 4D continuum has emerged. Defect-core peak $d_S = 9.06$: Causal Prisms trap walkers—mass is topological probability retention.

Mass hierarchy. $4.56 < 6.54 < 7.73$ (ratio 1 : 1.43 : 1.69): higher-generation prisms trap walkers more strongly due to more complex phase structures.

CPT symmetry. Gen 1 = 4.56 vs Anti-1 = 4.45 ($\Delta m/m = 2.4\%$ at $M=2$), expected to sharpen under larger ensemble averaging.

THE THREE GEOMETRIES

Geometry of Order. Poisson sprinkling + transitive reduction yields the Hasse diagram [5, 14]. It is triangle-free by construction: if $u \prec w \prec v$, the direct link $u \rightarrow v$ is removed. Therefore K_4 cannot appear as a subgraph of the Hasse diagram. The densest bipartite structure

is $K_{2,n}$ —the *Causal Prism*: a past pole u and future pole v connected through $n \geq 3$ mutually spacelike (disconnected) intermediates [10, 15]. Topological mass = n .

Geometry of Matter (Generation Theorem). Each intermediate carries a discrete phase: $\varphi(w_i) = \text{sgn}(p_{\text{bulk}}[w_i]) \in \{-1, 0, +1\}$. The generation of a prism is

$$g(P) = |\{\text{distinct } \varphi(w_i)\}| \in \{1, 2, 3\}. \quad (4)$$

Exactly three, because the sign function has three values—a theorem, not a fit. The net phase $\Phi(P) = \sum_i \varphi(w_i)$ distinguishes matter ($\Phi \geq 0$) from antimatter ($\Phi < 0$).

K_5 threat absorption enforces confinement: an isolated prism component would create a forbidden K_5 topological minor [10], so prism constituents cannot be separated—mirroring colour confinement in QCD.

The S_n permutation symmetry of the n intermediates (the “belly” of the prism) is not a hint of gauge structure—it *is* the gauge structure. For $n = 3$ intermediates, S_3 is the Weyl group of $SU(3)$: the colour charge algebra of QCD derived from the bipartite combinatorics of the Hasse diagram.

Geometry of Logic. $D \leq 15 \Rightarrow O(N)$. This bound is not imposed—it is a geometric consequence of $\rho \approx 1$ sprinkling into the Hasse diagram of a causal diamond. The simulation’s $O(N)$ efficiency is the computational manifestation of physical locality [6, 16].

MACROSCOPIC EMERGENCE

Gravity as information cost. By Chentsov’s theorem [8], the unique metric on the space of causal histories is the Fisher information metric: $g_{\mu\nu} \propto I_{\mu\nu}(\theta)$. Gravity is the statistical resistance of the causal network to information deformation. As Jacobson showed [17], when causal flux is bounded by a horizon, $S \propto A/4G$ and the Einstein equations emerge as a thermodynamic equation of state [18, 19].

Dark energy as Poisson noise. Volume fluctuations $\delta V \sim \sqrt{N}$ [20] yield

$$\Lambda \propto \frac{1}{\sqrt{N}} \propto H^2 \quad (5)$$

Dark energy is the statistical jitter of counting causal events; it is tiny because $N \gg 1$ [21].

IMPLICATIONS

By strictly adhering to the Kuratowski Calculus, major anomalies dissolve as artefacts of the continuum assumption:

1. **Singularity:** Holographic capacity bounds Planck volumes to 1 bit; infinite density is obstructed.
2. **Dark energy:** Poisson noise, not vacuum energy (Eq. 5) [20].

3. **Proton stability:** Prisms are topological knots; decay requires reordering $\sim 10^{60}$ links.
4. **Strong CP:** θ is the average of $\sim 10^{60}$ random link phases; by the CLT, $\theta \rightarrow 0$.
5. **Baryogenesis:** Forward-growing DAGs favour $\Phi > 0$ (matter) over $\Phi < 0$ (antimatter); the simulation confirms $\sim 13:1$.

FALSEIFIABLE PREDICTIONS

The framework is an *overconstrained system*: zero free parameters, seven independent predictions. Every observable is coupled to every other through the belly-size distribution $f(N)$ of Causal Prisms. There is nothing to tune; failure of any single prediction is fatal.

P1. Generation Exclusion ($g_{\max} = 3$). The generation number $g = |\{\text{distinct } \varphi(w_i)\}|$ takes values in $\{1, 2, 3\}$ because $\text{sgn}: \mathbb{R} \rightarrow \{-1, 0, +1\}$. A fourth generation is topologically forbidden—a theorem, not a bound.

P2. Neutrino mass sum ($\sum m_\nu \approx 58\text{--}62\text{ meV}$). Bulk diffusion gives $m_1 \approx M_{\text{Pl}}(\ell_P/L_\Lambda)^{1/2} \approx 3\text{ meV}$. Combined with measured Δm^2 splittings: $\sum m_\nu \approx 62\text{ meV}$, normal ordering. Testable by DESI + Euclid + CMB-S4 lensing ($\sigma \sim 20\text{ meV}$).

P3. No primordial tensor modes ($r < 10^{-3}$). UV dimensional reduction $d_S \rightarrow 2$ kills graviton propagation (Weyl tensor vanishes in 2D). Distinguishes from slow-roll inflation ($r \sim 10^{-2}\text{--}10^{-1}$). Testable by CMB-S4 and LiteBIRD.

P4. The α - Ω Lock. The topological charge ratio $\mathcal{Q}_{\text{topo}} = \sum |\Phi|^2 / \sum N^2$ determines *both* observables: $\alpha = \mathcal{Q}_{\text{topo}}/(8\pi)$ and $\Omega_{\text{energy}} = 1/\mathcal{Q}_{\text{topo}} - 1$ (the self-energy dark matter ratio). The exact identity $\alpha(1+\Omega) = 1/(8\pi)$ locks them: one cannot vary dark matter without altering α . With $\mathcal{Q}_{\text{topo}} \approx 0.190$: $\alpha^{-1} \approx 132$ (within 3%), $\Omega_{\text{energy}} \approx 4.25$ (within 21% of Planck 5.36). The residual gap quantifies the baryonic confinement contribution ($K_{3,3}$) not yet modelled.

P5. Null WIMP prediction ($\sigma_{\chi N} = 0$). Dark matter is the phase-cancelled ($\Phi = 0$) tail of $f(N)$, not a new particle species. Cross-section is identically zero. All direct detection experiments (LZ, XENONnT, DARWIN) will report null.

CONCLUSION

From two axioms and one principle, the $N = 10^7$ simulation produces: (i) three generations bounded by sign trichotomy $g \leq 3$ —a theorem; (ii) mass hierarchy $4.56 < 6.54 < 7.73$ without calibration; (iii) CPT to 2.4%; (iv) $\Omega_{\text{energy}} = 1/\mathcal{Q}_{\text{topo}} - 1 = 4.25$ via self-energy decomposition (locked to α by $\alpha(1+\Omega) = 1/(8\pi)$, zero free parameters); (v) $\alpha = \mathcal{Q}_{\text{topo}}/(8\pi)$ with $\mathcal{Q}_{\text{topo}} = \sum |\Phi|^2 / \sum N^2 \approx 0.19$ giving $\alpha^{-1} \approx 133$ (emergent, zero

free parameters); (vi) $d_S = 2 \rightarrow 4$ consistent with CDT. $D \leq 15$ makes every operation $O(N)$: the code proves locality. Open: Yang–Mills from S_n ; physical mass units. All code and data: https://github.com/ertwro/fractal_entropic_geometrodynamic.

I dedicate this work to my wife and daughter. I acknowledge the computational endurance of a ThinkPad T480 running Arch Linux, and thank the AI architectures Claude, Grok, and Gemini for serving as adversarial semantic mirrors throughout the synthesis of this framework.

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